

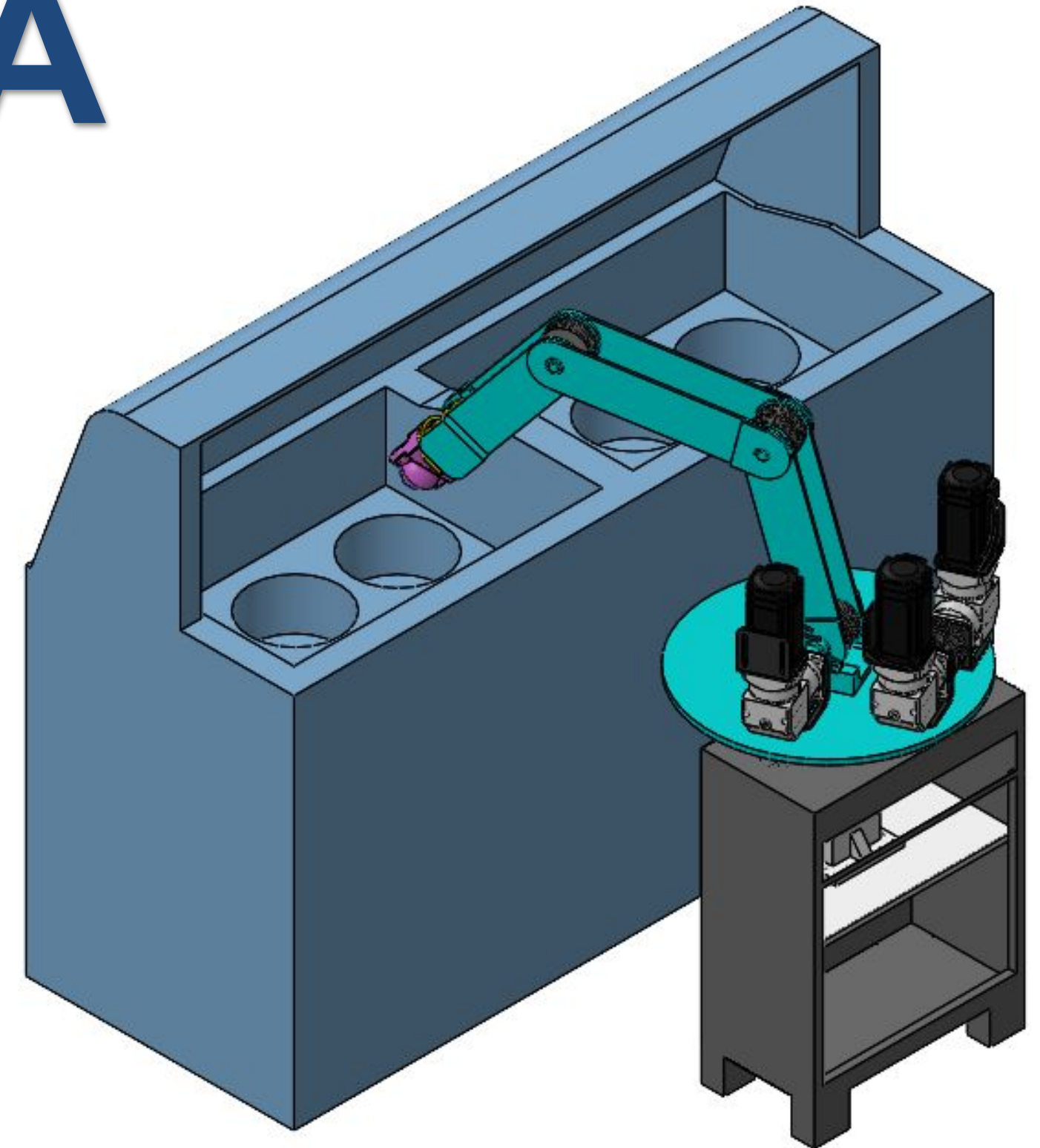
September 2026

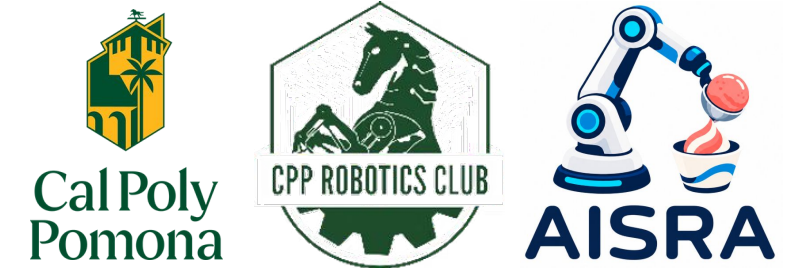


PROJECT AISRA

Autonomous
Intelligent
Scalable
Robot
Arm

Krishiv Parikh, Youssef Magdy, Rayyan Hisham, Hunter Heflin,
Daniel Sandoval, Christopher Tancher, Isaiah Renteria



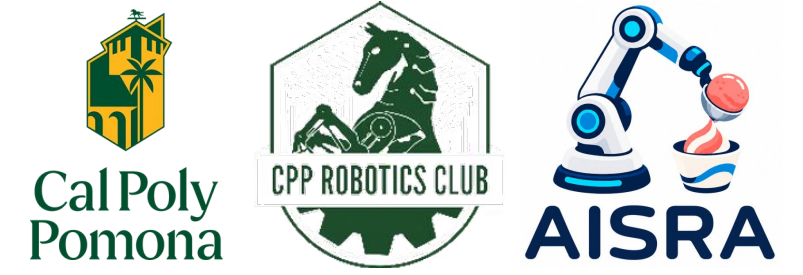


The Project - AISRA

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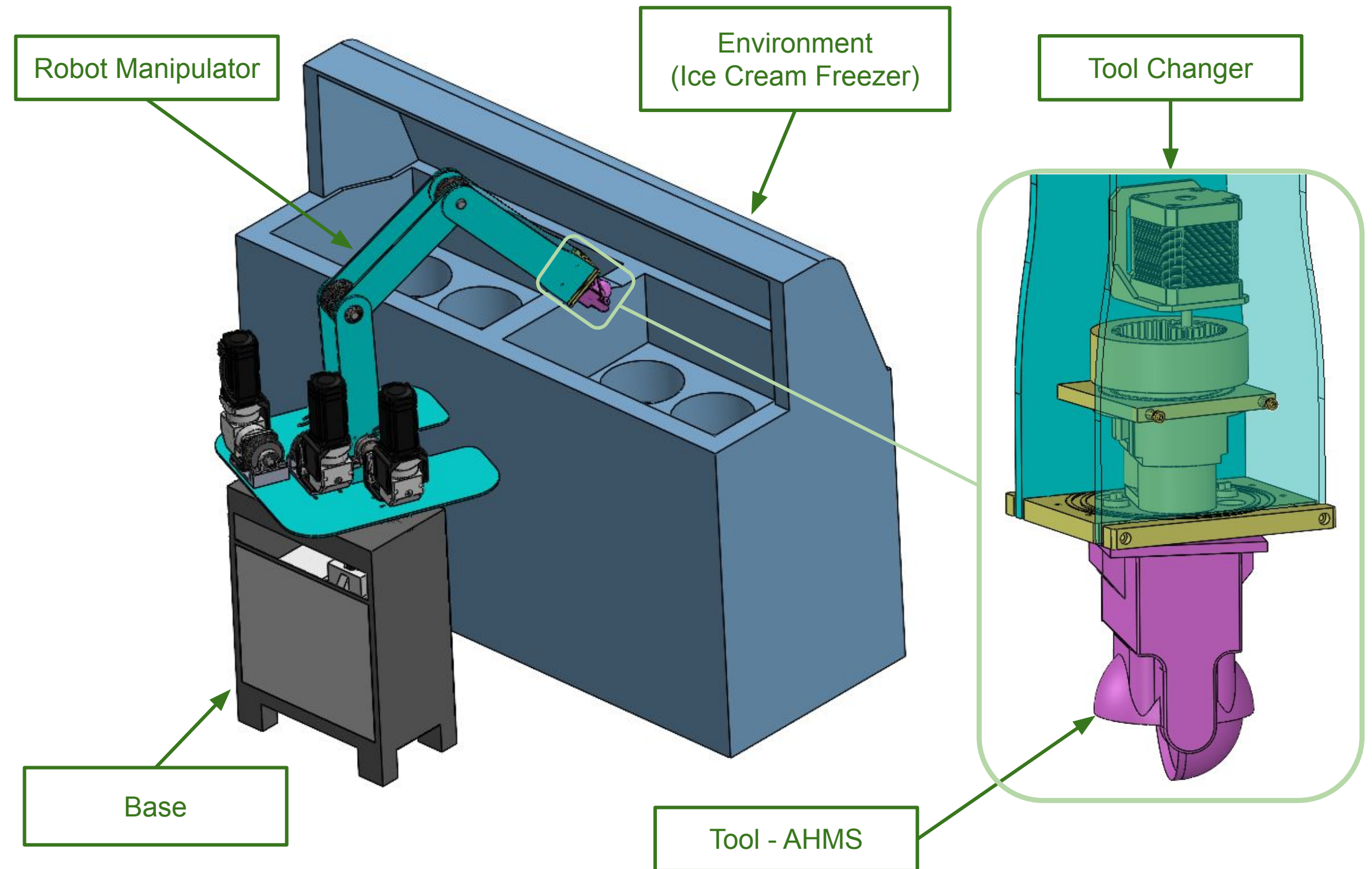




The Project - AISRA

Overview Autonomous Intelligent Scalable Robot Arm (AISRA)

- Our team is developing AISRA, a low-cost alternative to conventional industrial robotic arms.
- Designed as a scalable solution for diverse automation tasks.
- Tasked to scoop ice cream with a custom, detachable end-effector (Project AHMS)



** Minor components of AISRA (ex. pulley cables, electronics) are hidden for clarity. Additionally, the manipulator will be fully covered with a thin 3D printed shell to keep debris out of our system.

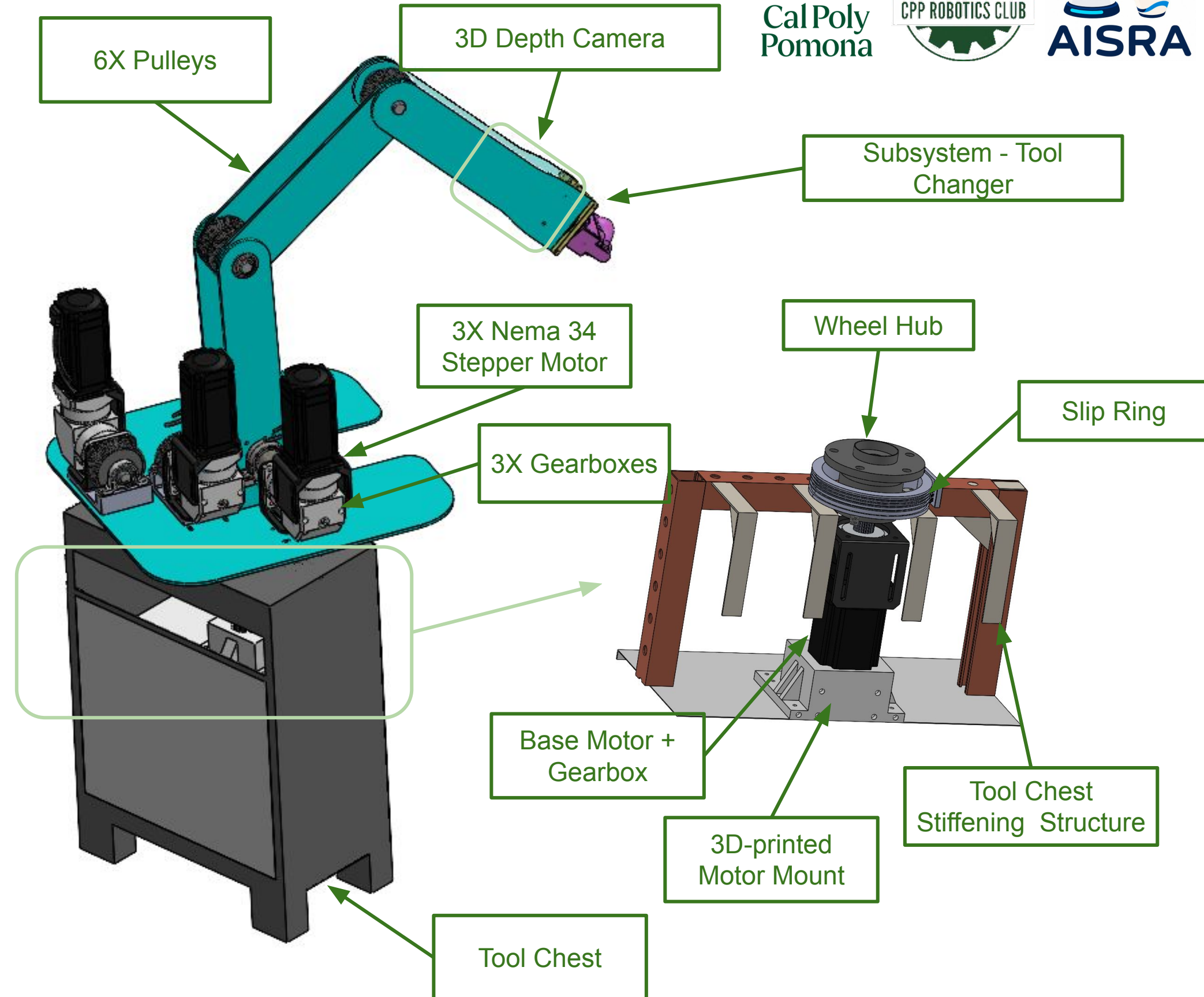
Robot Manipulator

Function

Provides five-axis robotic manipulation and autonomous tool (end-effector) changing as a lower-cost alternative to conventional industrial robot arms.

Overview

- Manipulator Specifications
 - Acceleration: 3.14 rad/s^2
 - Payload: 4.5kg (~10lbs)
 - 1.2 meter maximum reach
- Cable driven actuation for arms 2 and 3
- Direct driven architecture for arm 1 base and effector revolution
- Powered using Nema 34s + gearboxes to suit torque requirements
- Custom slip ring to enable continuous 360° rotation of turntable
- Turntable powered via wheel hub bearing connected to Nema 34 & gearbox
- Tool chest fitted with structural enhancements to support robot arm at max load cases
- 3D depth camera for environment mapping



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Subcomponent - Tool Changer

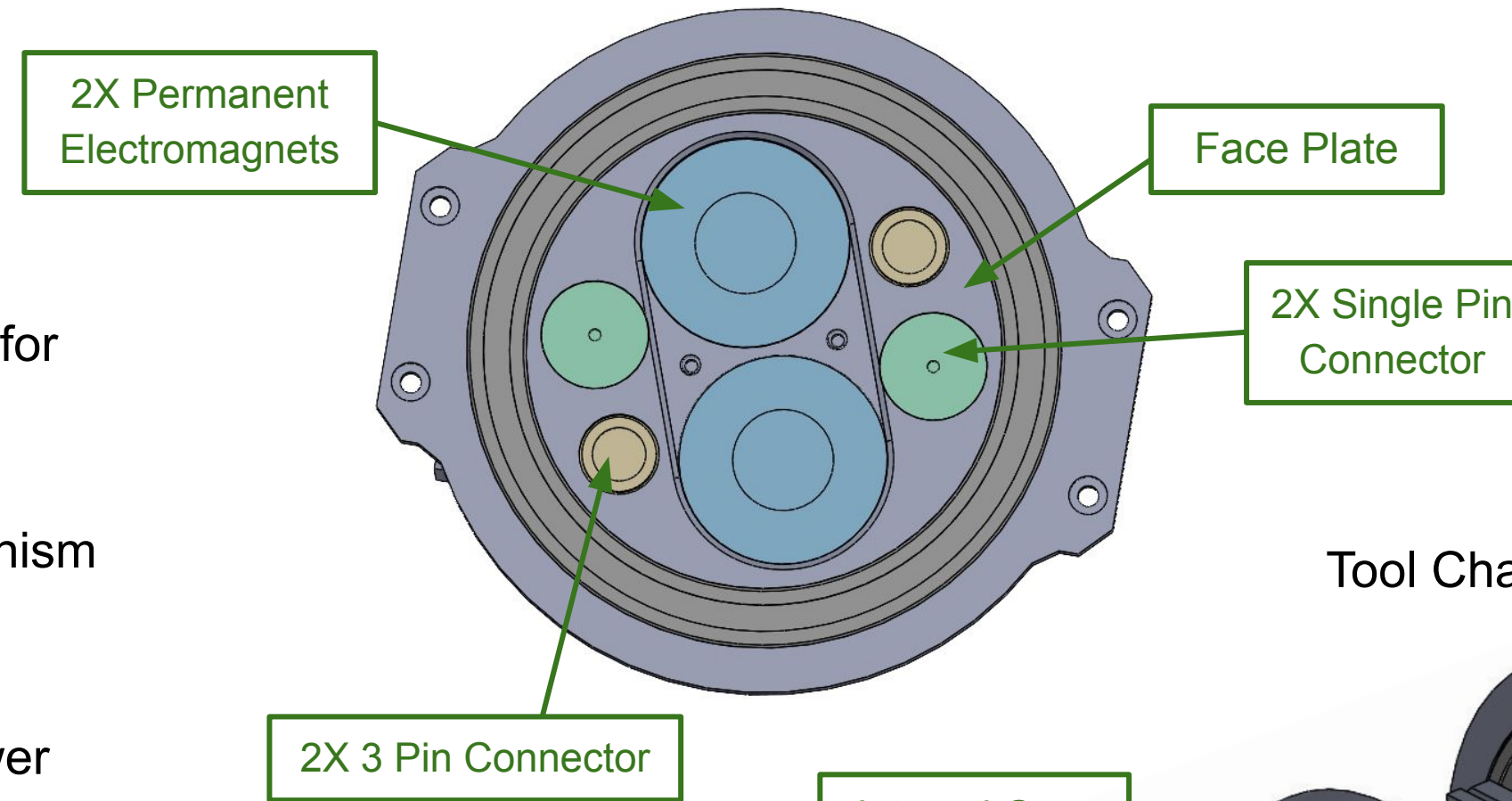
Function

Provides power, bi-directional communication, securement, and continuous rotation to the tool.

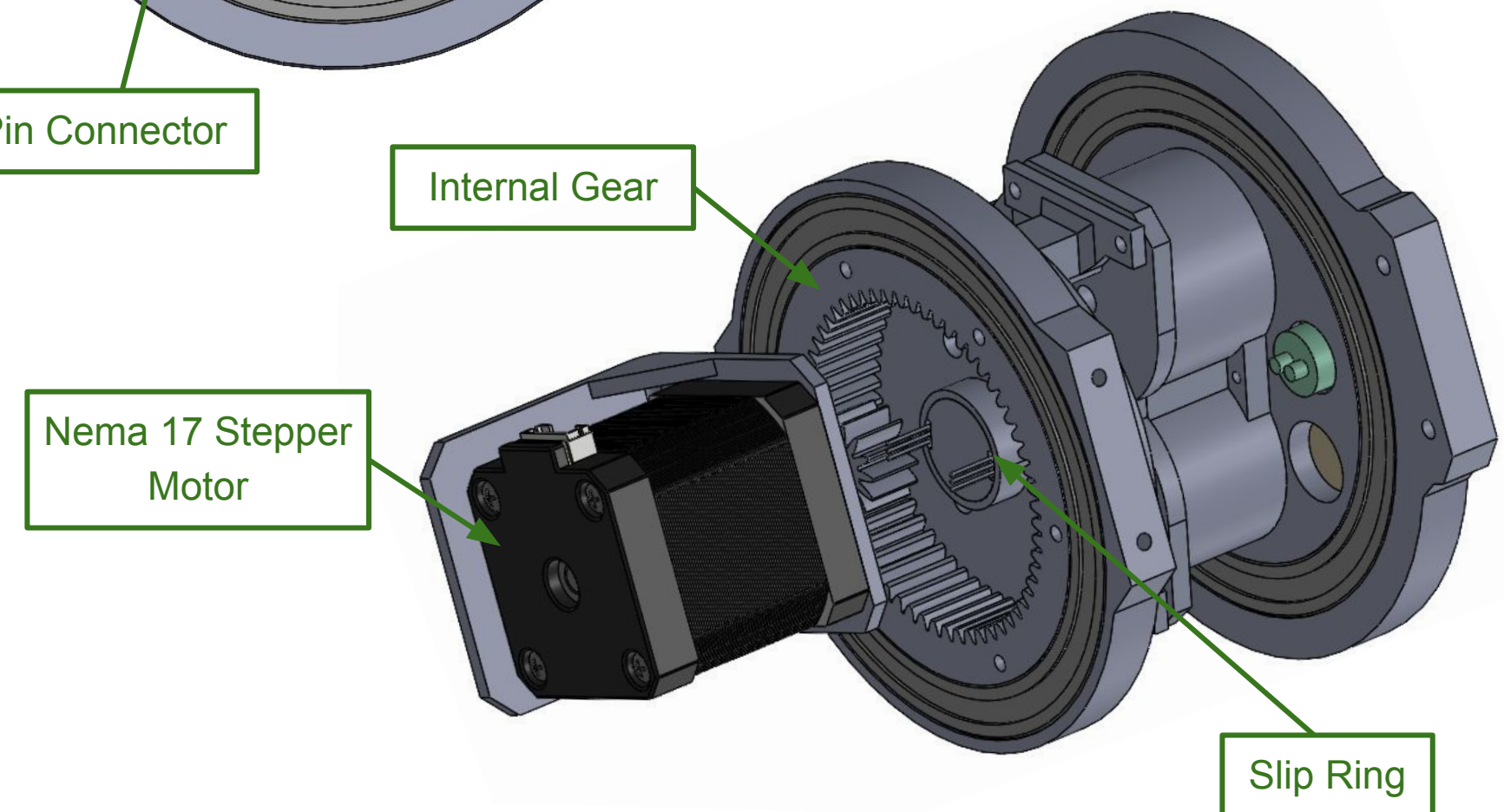
Overview

- Rotationally symmetric face plate design to allow for dual-orientation tool mounting
- Slip ring to enable continuous 360° rotation
- Nema 17 stepper motor with internal gear mechanism for upto 3 N·m output torque
- Features electro-permanent magnets
 - Enables secure tool coupling without any power draw
 - Counteracting magnetic field allows controlled tool release
 - 90 N holding force (~20 lbf)
 - System rated for 10 lbf, providing a 2× safety factor
- Single pin connector for upto 10 A power delivery
- Three pin connector for TX, RX, Ground
- 1 machined faceplate. Remaining components are OTS or 3D-printed with nylon carbon fiber filament

Tool Changer - Top Down View



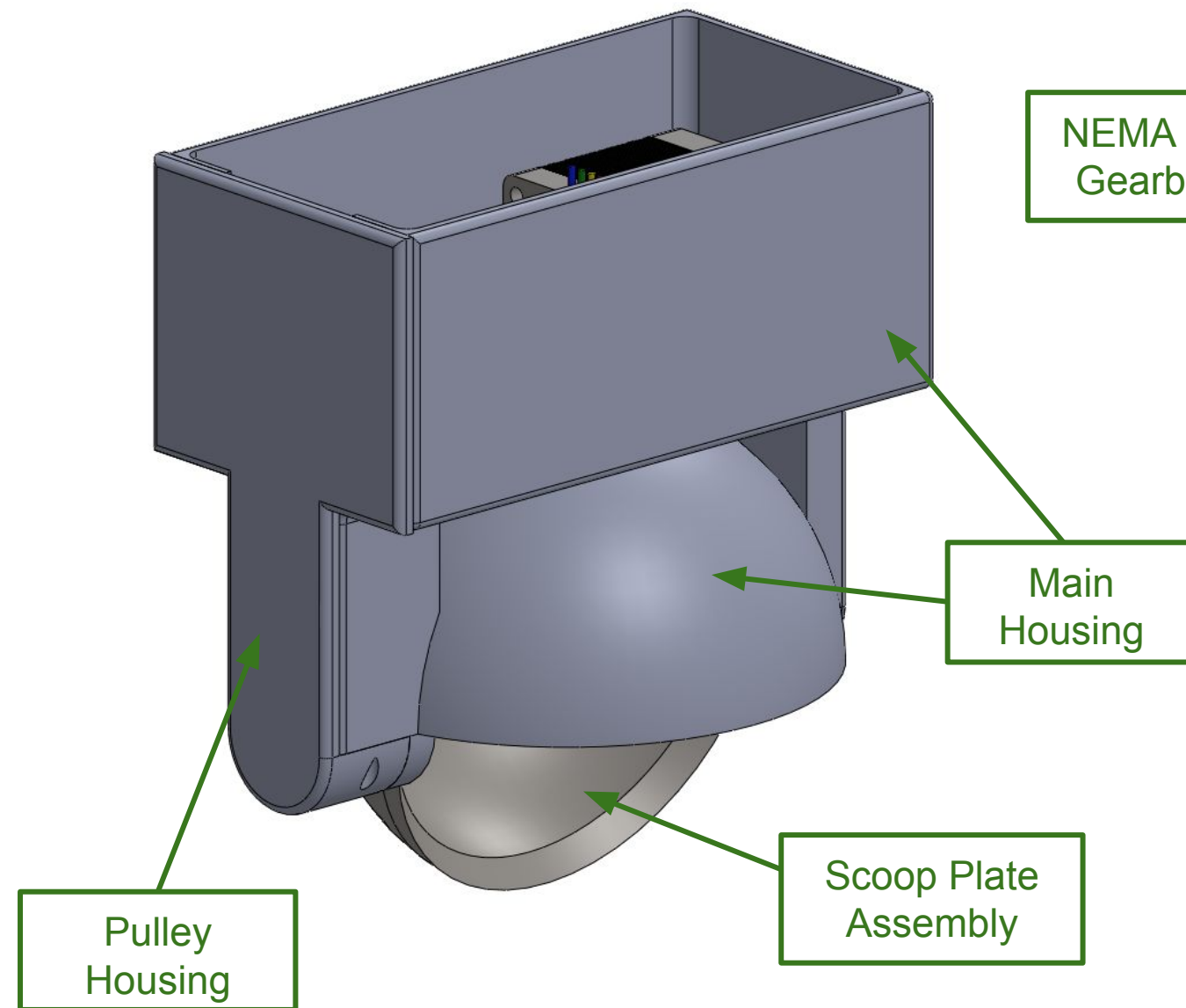
Tool Changer - Isometric View



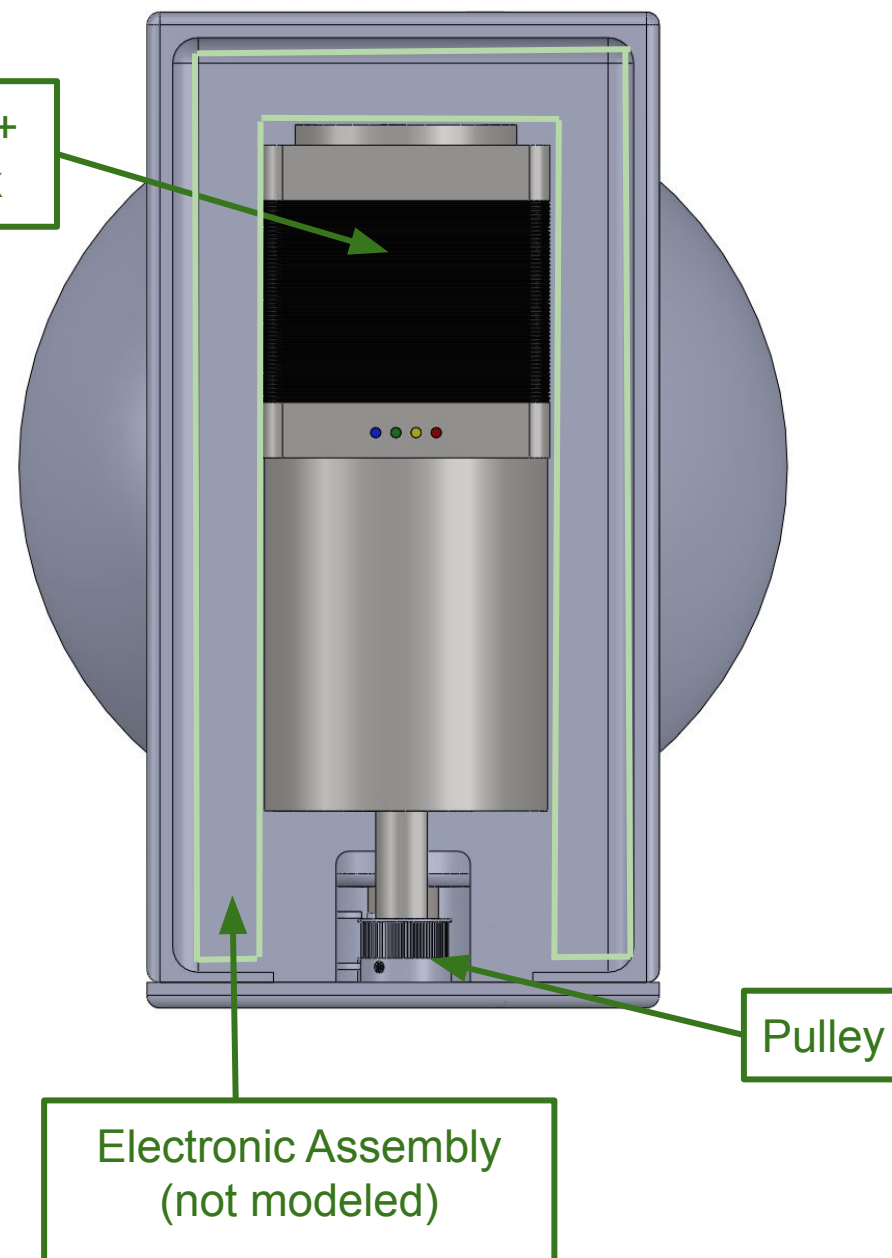
Tool - AHMS

Autonomous Heated electro-Mechanical Changer

AHMS - Isometric View



AHMS - Top Down View



Function

Scoops a 4 oz serving of ice cream using a heated, rotating scoop plate while maintaining the desired, spherical scoop form.

Overview

- NEMA 8 stepper motor and gearbox drive the scoop plate through a cable-and-pulley mechanism
- Integrated heating element housed in the scoop plate assembly raises the scoop plate surface to 180°F
- Thermistor feedback maintains the target operating temperature
- Heated scoop plate reduces cutting resistance and slices through the ice cream
- Approximate heat-up time of 5 seconds
- Electronic and drive components are enclosed above the food-contact region

**Top plate that attaches to the tool changer, and a few other components (ex. fan, circuit board) are hidden from both views for clarity.

Prototyping - Manipulator

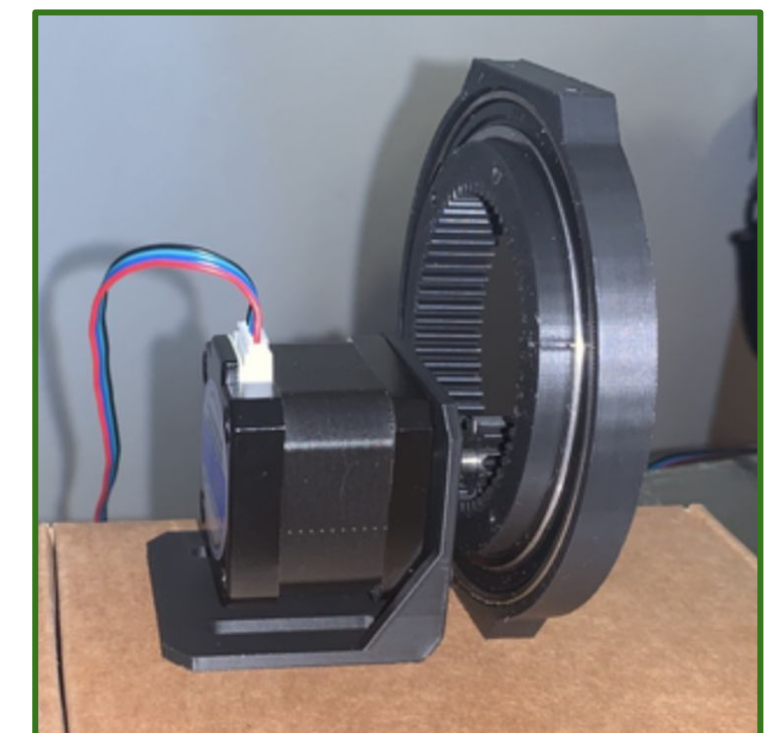
We have completed two prototypes of the robotic manipulator. The first validated the belt-driven architecture, while the second helped us characterize the drivetrain dynamics and gain experience with our initial base-drive design through system integration. The tool changer is currently in the prototyping phase.



Robot Manipulator Prototype 1 Video
([Link 1](#))



Robot Manipulator Prototype 2 Videos
([Link 1](#), [Link 2](#))



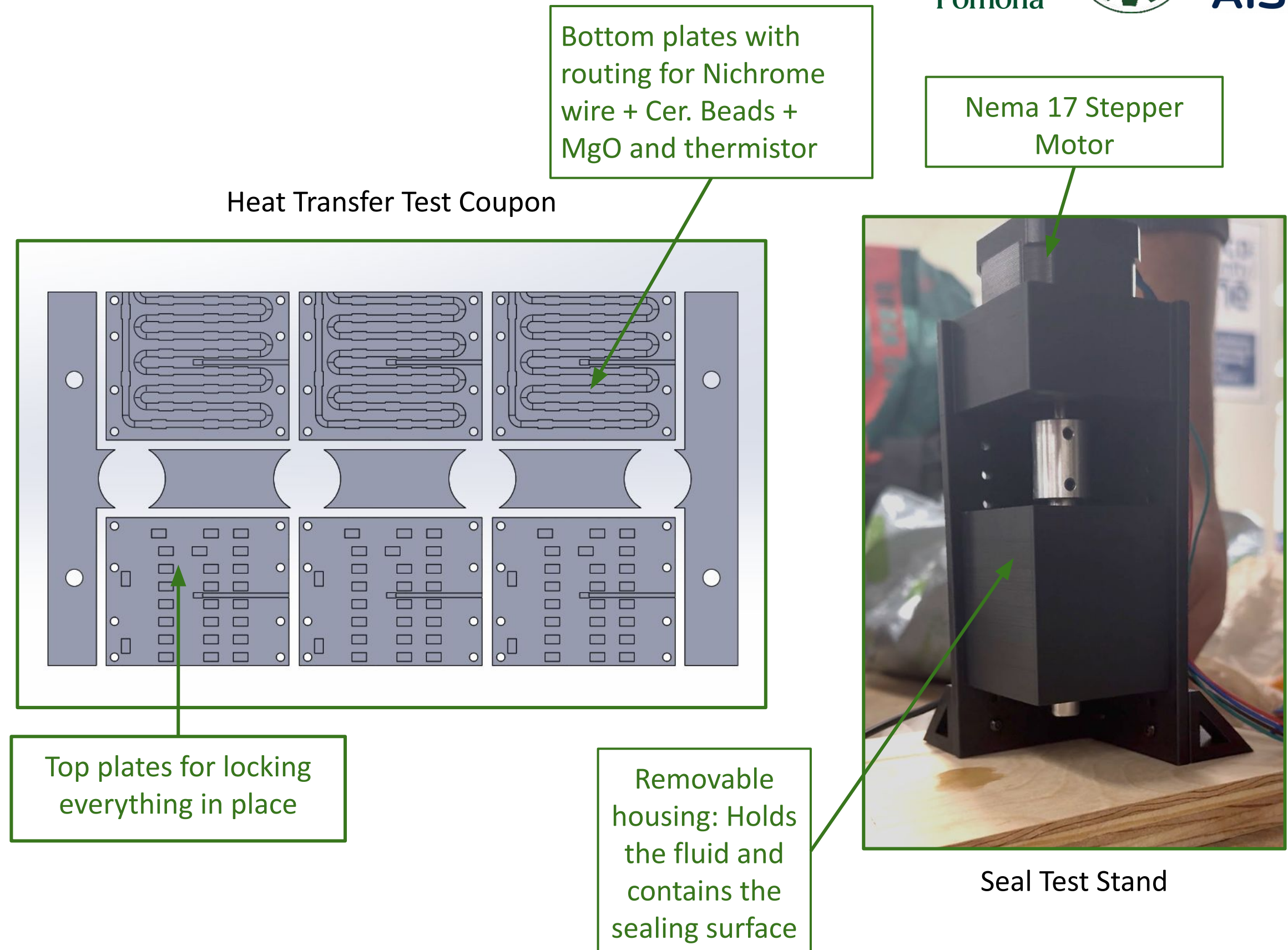
Tool Changer Prototyping
Photos (Ongoing)

Prototyping - AHMS

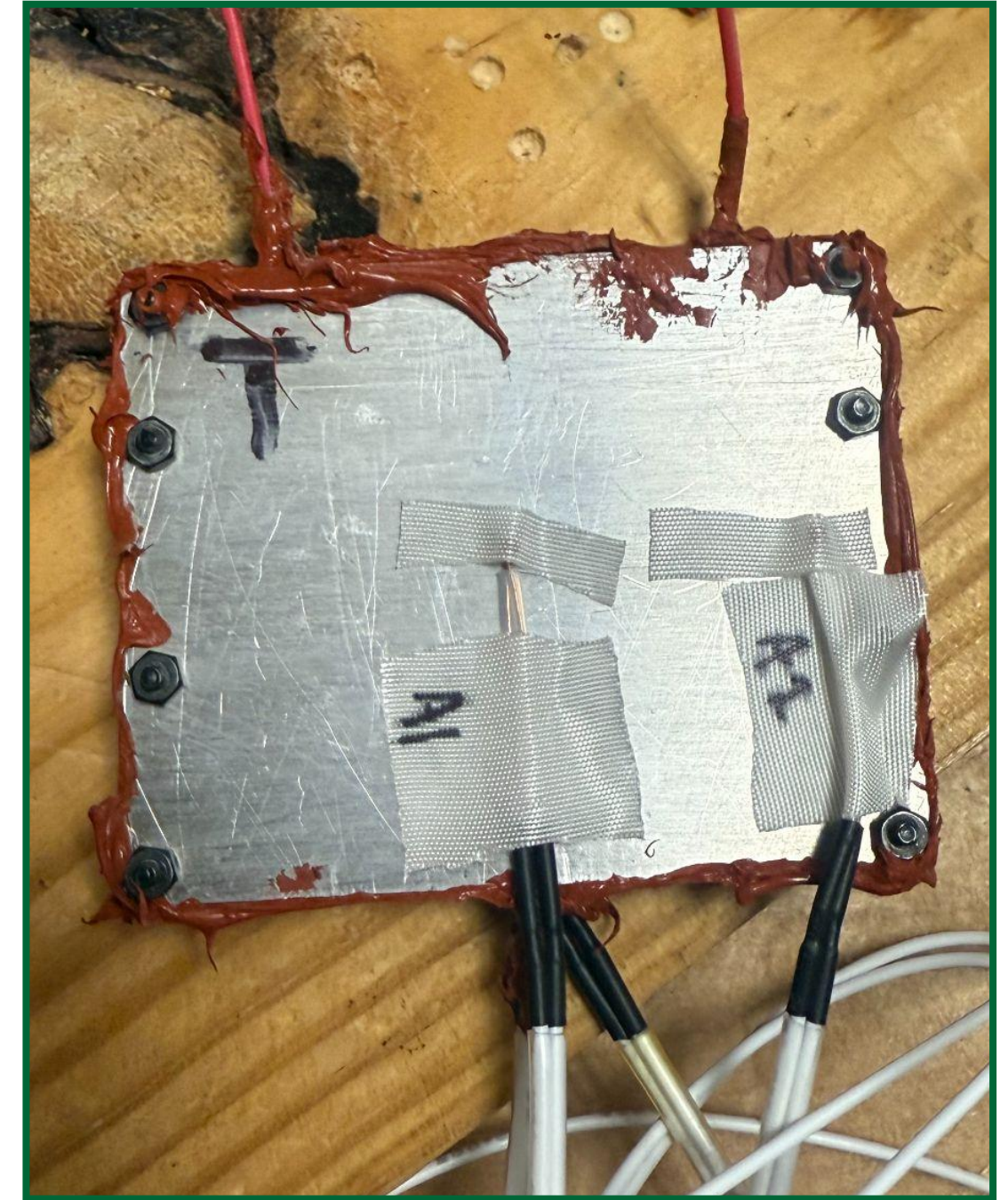
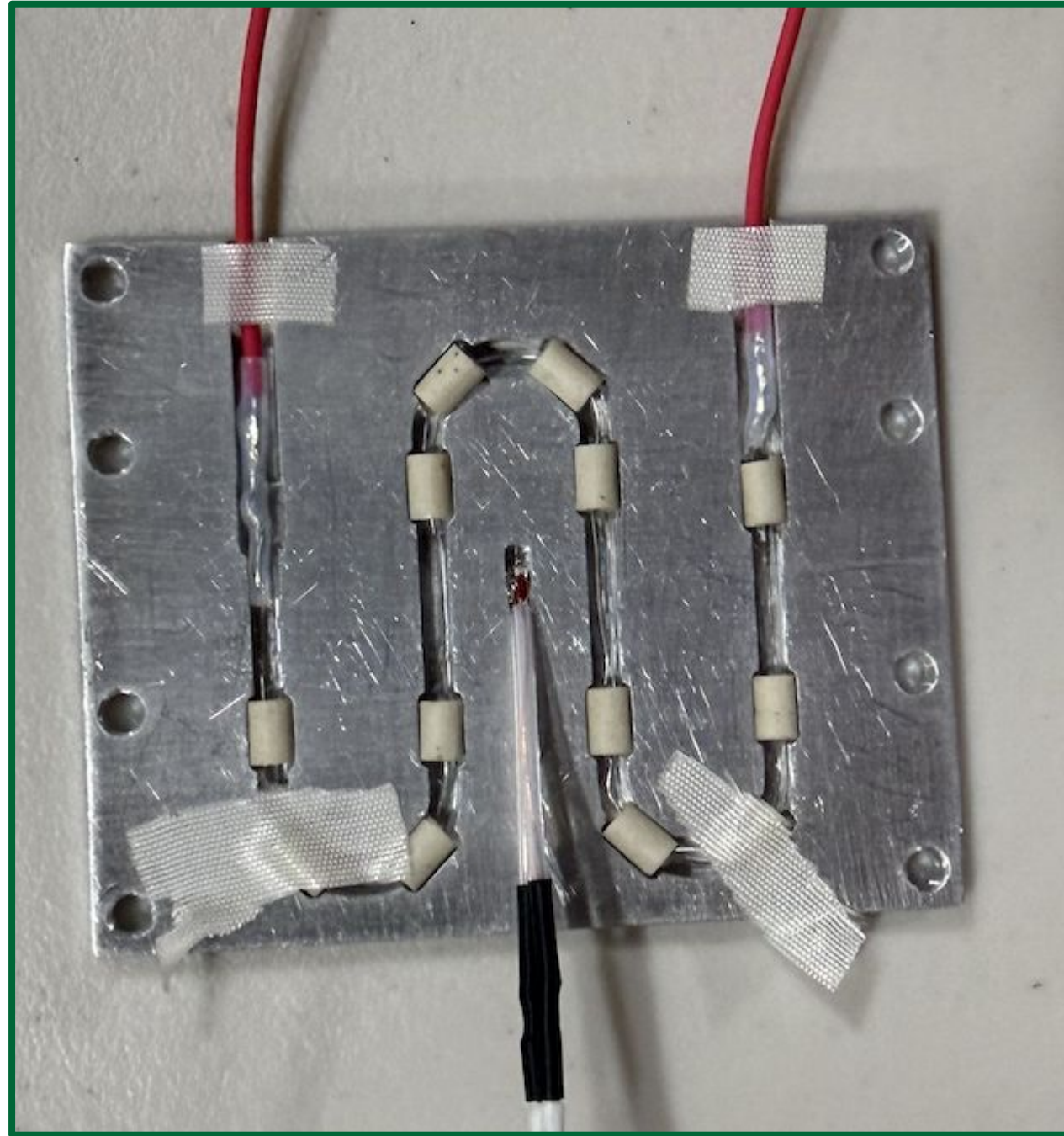
Autonomous Heated Electro-Mechanical Changer

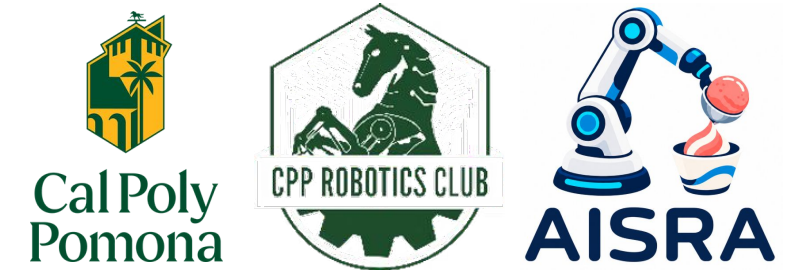


- Because the design requires food-safe, heat-resistant materials, full 3D-printed prototypes aren't viable. Instead, we tested individual components separately to verify the design.
- Seal performance was tested in cold (0 °C) and hot water (89 °C).
 - Both mechanical and V-seals were tested across various speeds and durations.
 - The mechanical seal better accommodated shaft misalignment and passed both tests, so it was selected.
- Heating-element testing is underway to validate heat-transfer calculations and assembly viability.
 - The assembly uses machined 6061 aluminum with custom element routing and a temporary silicone moisture seal.
 - More photos of it on the next slide



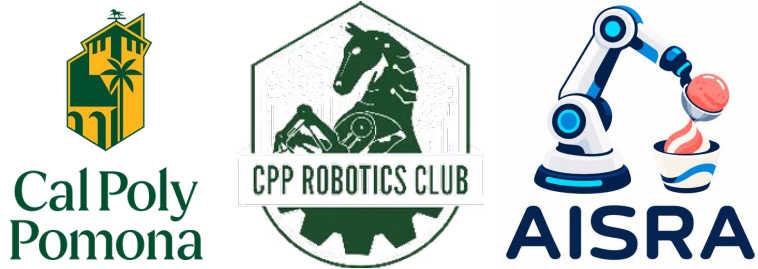
Prototyping - AHMS





Team Information

Timeline & Team Structure - August 2026



Timeline

Past Work....	August	September (Current)	October
• AHMS heating mechanism research, testing design • AHMS main housing and drive system design • Tool changer design • Manipulator prototypes • Final Manipulator structural analysis and design • Robot motion: inverse kinematics, closed loop control from sensor feedback • Initial scooping robot operational logic algorithm • Heating element test circuit board	• Tool changer prototyping and testing • Manipulator base construction • Scooping mechanism drive system testing • Integrating feedback controls into prototype arm • Robot motion testing and iteration on prototype manipulator • Tool changer circuit board	• Scooping mechanism DFM and manufacturing • Manipulator manufacturing • Robot motion testing and iteration on prototype manipulator • Scooping mechanism circuit board	• Manipulator manufacturing and assembly • Electronics integration • Robot motion testing and iteration on prototype manipulator • Scooping mechanism manufacturing and integration
		• November	• December
		• Robot motion integration & tuning - final manipulator • Integrate and iterate robot operation logic algorithms • Tune scooping mechanism PID parameters	• Project Showcase

Project Structure

Project Lead
Krishiv Parikh

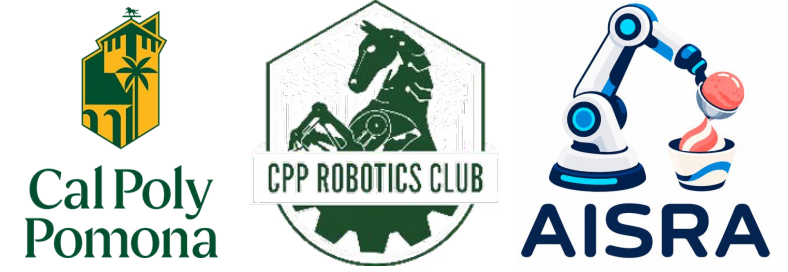
Mechanical Team - Manipulator
Rayyan Hisham (Lead), Krishiv Parikh,
Daniel Sandoval, Isaiah Renteria

Mechanical Team - AHMS
Youssef Magdy (Lead), Hunter Heflin

Controls Team
Jinsung Choi, Krishiv Parikh

Electrical Team
Christopher Sanchez

Questions?



- Interested in joining or have any other questions?
 - Email the following:
 - Krishiv Parikh - Projects Lead Engineer, krishivparikh09@gmail.com
 - Rayyan Hisham - Manipulator Lead Engineer, rayyanhisham@outlook.com
 - Youssef Magdy - AHMS Lead Engineer, youssef.magdyzd@gmail.com